The S_3-symmetric 3HDM fermion sector: three corrections to the LFVHD draft, and why exact S_3 forces $V_{us} = 0$

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Abstract

I have rebuilt the fermion sector of our LFV draft [1] from the Lagrangian, using the draft's own S_3 assignment, and checked every derivation in it. The charged-lepton Yukawa sector is invariant and complete. Three defects turned up, and all three are now corrected in `LFVHD_3HDMS3.tex`. The one with consequences is that $\mu_5 = \mu_4$ is an *assumption*, not a result: it follows from matching masses to eigenvalues rather than singular values. The draft does not treat quarks, so I added them, and this gives the substantive result. With exact S_3 the CKM matrix is forced to be a pure 2–3 rotation, $V_{us} = V_{ub} = V_{cd} = V_{td} = 0$, for *any* Yukawa couplings. The cause is the same residual $\mathbb{Z}_2$ that the vacuum alignment preserves. Soft S_3 breaking lifts it: a bounded fit then reproduces the six quark masses and $|V_{us}|$, $|V_{cb}|$, $|V_{ub}|$ simultaneously. The CP phase stays out of reach, because these Yukawas are real.

1 Summary

Everything below comes from `research/thdm_s3/02_s3_fermion_sector.ipynb`. It takes the S_3 assignment, the Yukawa Lagrangian and the diagonalisation strategy from [1]. Everything else is derived with `feynlag`, including every check of the draft.

1. **The draft and our code use different S_3 doublet bases**, related by a reflection at $\pi/6$. For the Yukawas the coupling dictionary comes out as the identity. This contrasts with the scalar potential, where the same analysis flips the sign of λ_4 (§2).
2. **The lepton Yukawa sector is invariant and complete**: five structures, all gauge- and S_3-invariant, and an independent enumeration finds exactly five (§3).
3. **Three defects in the draft**, all corrected in the `.tex`. One needs further wording in the draft: $\mu_5 = \mu_4$ should be stated as an assumption (§4).
4. **Exact S_3 forces $V_{us} = 0$** (§5), which is the main result.
5. **Soft S_3 breaking switches the Cabibbo angle on**, at no cost to the scalar constraints (§6).

2 The two bases

The draft's doublet generators are $a = R(-2\pi/3)$ and b, a reflection about 30°. `feynlag`'s are $\rho(a) = R(2\pi/3)$ and $\rho(b) = \mathrm{diag}(1, -1)$. The two are conjugate by a unique $O(2)$ map, the

reflection at $\pi/6$:

$$O = \begin{pmatrix} \cos\frac{\pi}{6} & \sin\frac{\pi}{6} \\ \sin\frac{\pi}{6} & -\cos\frac{\pi}{6} \end{pmatrix}, \qquad O^{-1}\rho(g)\,O = \rho_{\text{draft}}(g), \qquad \det O = -1.$$

This is why the draft's alignment reads $v_1 = \sqrt{3}\,v_2$ while ours reads $v_2 = \sqrt{3}\,v_1$. Of the five Yukawa structures, four are dot products of two doublets and do not depend on the basis. Only the triple-doublet structure (Y_2) changes. Each form is invariant in its own basis and fails in the other. Under O the draft's form maps to ours with coefficient $+1$, so the coupling dictionary is the identity.

The vacuum needs one remark. The map sends our aligned vacuum $(v_1, \sqrt{3}v_1)$ to $(\sqrt{3}v_1, -v_1)$, which is the draft's $|v_1/v_2| = \sqrt{3}$ but with $v_2 < 0$. The sign carries no physics. The mass matrix obeys $M(-r) = P\,M(r)\,P$ with $P = \text{diag}(-1, 1, 1)$, which is a sign flip of the first-generation fields, so masses and |mixings| are unchanged.

3 The lepton sector is invariant and complete

With (L_1, L_2), (e_{1R}, e_{2R}) as doublets and L_S, e_{SR} as singlets [1], all five structures $Y_k^\ell T_k$ pass $SU(2)_L$, $U(1)_Y$ and S_3 invariance, and the Lagrangian with its h.c. is hermitian. The draft's literal T_2 fails S_3, as it must, because it is written in the other basis. `feynlag`'s own enumeration of S_3- and gauge-invariant Yukawa operators, run independently of the draft, returns exactly five. The charged-lepton Yukawa sector of [1] is complete.

4 Three defects in the draft, all corrected

1. The $\mu \leftrightarrow Y$ dictionary was a factor $\sqrt{2}$ too large. The draft writes an explicit $1/\sqrt{2}$ on the Yukawa structures and also expands $\langle H_i^0 \rangle = v_i/\sqrt{2}$. Composing the two gives, for example, $\mu_1^\ell = Y_1^\ell v_3/2$, where the draft printed $v_3 Y_1^\ell/\sqrt{2}$. All five printed μ_k exceed the Lagrangian's by exactly $\sqrt{2}$, which makes it a convention slip ($\langle H^0 \rangle = v$ instead of $v/\sqrt{2}$) and not a typo. Nothing physical depends on it, since the μ's are eliminated in favour of the masses. It matters only if one quotes the Yukawa couplings themselves.

2. Eigenvalues versus singular values; $\mu_5 = \mu_4$ is an assumption. After O_{12} the residual block is $A = \begin{pmatrix} \rho & 2\mu_5 \\ 2\mu_4 & \mu_3 \end{pmatrix}$ with $\rho = \mu_1 + 2\mu_2$. The physical masses are its *singular* values. From $\sigma_1^2 + \sigma_2^2 = \text{tr}(A^\mathsf{T} A)$ and $\sigma_1\sigma_2 = \det A$,

$$(\sigma_1 - \sigma_2)^2 = (\rho - \mu_3)^2 + 4(\mu_4 + \mu_5)^2,$$

while the draft printed $+16\mu_4\mu_5$. That is $(\text{tr}A)^2 - 4\det A$, the gap between the *eigenvalues*. The two agree only when $\mu_4 = \mu_5$. More important, $\mu_5 = \mu_4$ is not forced by the masses. It is equivalent to imposing the trace relation $m_\mu + m_\tau = \text{tr}A$, i.e. to the ansatz $O_L = O_R$. The notebook exhibits a $\mu_4 \neq \mu_5$ point with valid masses as singular values (singular values 2.539, 0.291; eigenvalues 2.232, -0.332; $\text{tr}A = 1.9$). The corrected `.tex` fixes the formula, but it still uses the trace relation, so **the draft should state $\mu_5 = \mu_4$ as an assumption**. This is the one wording change still open.

3. A coefficient slip. In the $\mu_5 = 0$, $\mu_3 = m_\mu$ case, $\text{tr}(A^\mathsf{T} A) = m_\mu^2 + m_\tau^2 + 4\mu_4^2$, where the draft printed $2\mu_4^2$. The conclusion $\mu_4 = 0$ is unaffected.

What checks out. Everything else in the draft's fermion sector agrees with an independent derivation:

- $R_S = R_A R_H$ holds identically;
- O_{12} block-diagonalises the 1–2 sector for *any* vacuum ratio, and at the alignment the first generation decouples with $m_e = \mu_1 - 2\mu_2$;
- $\tan 2\theta_\ell = -4\mu_4/(m_\mu + m_\tau - 2\mu_3)$ and the $p_{1,2}$ closed forms agree;
- $Q_1(A) = \mathrm{diag}(m_e, m_\mu, m_\tau)/v$.

The last is an identity, $Q_1(A) = O^\mathsf{T} M O/v$ for any orthogonal O. It pins the vacuum geometry rather than the rotation, so the G_k decomposition of M_ℓ is checked separately.

5 Main result: exact S_3 forces $V_{us} = 0$

The quark sector has ten Yukawa structures: five down-type through H and five up-type through $\tilde{H}$. All ten are S_3-, $SU(2)_L$-, $U(1)_Y$- and $SU(3)_c$-invariant. Because the S_3 matrices are real, $\tilde{H}$ is the same S_3 doublet as H, so M_u **and** M_d **have identical structure**. Only the couplings differ.

The 1–2 block-diagonalising rotation satisfies

$$\tan 2\psi = -r, \qquad r \equiv v_1/v_2 \ \ \text{(draft basis)},$$

and so depends on the *vacuum alone*, never on the Yukawas. The same O_{12} therefore acts on the up, down and charged-lepton sectors. The first generation decouples from the other two iff its eigenvector is orthogonal to $(r, 1)$, which reduces to

$$r\left(2 - \sqrt{r^2+1}\right) = 0 \qquad \Longleftrightarrow \qquad |r| = \sqrt{3} \ \ (\text{or } r = 0),$$

exactly the vacuum that preserves the residual $\mathbb{Z}_2$. With exact S_3 the tadpoles force that vacuum. So

$$V_{\mathrm{CKM}} = O_u^\mathsf{T} O_d = O_{23}(\theta_u)^\mathsf{T} O_{12}^\mathsf{T} O_{12} O_{23}(\theta_d) = O_{23}(\theta_d - \theta_u),$$

a pure 2–3 rotation, with $V_{us} = V_{ub} = V_{cd} = V_{td} = 0$ for arbitrary Yukawas. This is checked numerically to machine precision, and the root $\sqrt{3}$ is isolated: the decoupling entry is non-zero at $r = 2,\ 3/2,\ \sqrt{5},\ 1/2$. The form $O_{12}O_{23}$ uses the $\mu_5 = \mu_4$ ansatz, but $V_{us} = 0$ does not: at $|r| = \sqrt{3}$ the first generation decouples from both rows and columns either way.

The mechanism is the one of Das, Dey and Pal [2]: an unbroken $\mathbb{Z}_2$ gives an approximately block-diagonal CKM matrix. **The block, however, is different.** There the Cabibbo block stays free, and soft breaking generates only the small elements. With the draft's assignment the block is 2–3, so V_{us} itself vanishes, against a measured $|V_{us}| = 0.2243$ [3]. The exact-S_3 quark sector is excluded.

6 Soft S_3 breaking switches the Cabibbo angle on

Soft breaking means S_3-violating terms of dimension below four, which here are quadratics only. In the CP-conserving case the hermitian quadratics $\mu_{ij}^2 H_i^\dagger H_j$ have six real parameters, two of them S_3-invariant, leaving four soft ones: the **2** pair $(x_{11} - x_{22},\ x_{12} + x_{21})$ and $(H_S^\dagger H_{1,2} + \text{h.c.})$. Two consequences:

- **no quartic is touched**, so every boundedness and unitarity result of the scalar-sector report carries over unchanged;
- **the vacuum is freed**: the tadpole system stops being over-constrained, r is no longer pinned to $\sqrt{3}$, and the $\mathbb{Z}_2$ breaks.

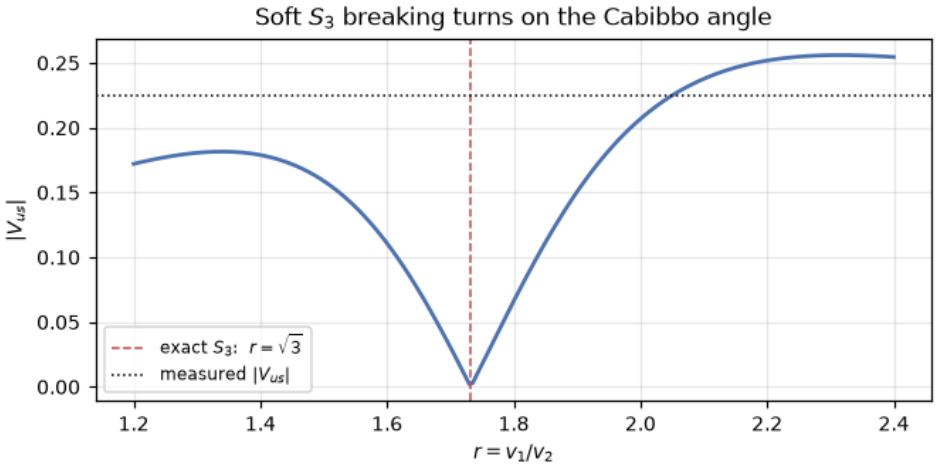


Figure 1: $|V_{us}|$ against the vacuum ratio r at fixed Yukawas. It vanishes exactly at the exact-S_3 vacuum ($|V_{us}| = 4.6 \times 10^{-17}$ at $r = \sqrt{3}$) and grows linearly away from it: the ratio of its values at $r = \sqrt{3} + 0.05$ and $\sqrt{3} + 0.025$ is 1.999.

To show that the mechanism reaches the observed values, I fitted the eight Yukawa parameters $(\mu_1, \ldots, \mu_4)_{u,d}$ plus r to the six quark masses and three CKM magnitudes [3]. The system has several exact solutions (cost below 10^{-20}; four with these targets). The one closest to the exact-S_3 value is kept, by that stated rule (cost 2.4×10^{-23}):

$$r = 1.3536 \quad (\text{exact } S_3 : \ \sqrt{3} = 1.7321), \qquad r - \sqrt{3} = -0.378,$$

reproducing all nine targets:

	$\lvert V_{us}\rvert$	$\lvert V_{cb}\rvert$	$\lvert V_{ub}\rvert$
fit	0.22431	0.04110	0.00382
target [3]	0.2243	0.0411	0.00382

together with the six masses,

$$(m_u, m_c, m_t) = (0.00216,\ 1.2730,\ 172.57)\ \text{GeV}, \qquad (m_d, m_s, m_b) = (0.00470,\ 0.0935,\ 4.183)\ \text{GeV}.$$

The fit is stored in `results/quark_soft_fit.json`.

Two caveats govern how to read this.

- **r is an existence proof, not a prediction.** Nine parameters against nine targets is an exactly determined system with several exact solutions at different r; which restart has the smallest cost is numerical noise, so the notebook lists all of them and selects by the rule above. What the fit establishes is that the two-stage structure does *not* prevent the three magnitudes from being matched simultaneously.
- **The CP phase is structurally out of reach.** The Yukawas are real, so the CKM matrix is real and the Jarlskog invariant vanishes identically.

Where this vacuum sits in the soft-broken *scalar* scan is the subject of the companion report on the soft-broken vacuum. The fit's $|r| = 1.354$ corresponds to $\varphi = 53.5°$ inside the scanned domain, a well-populated region.

7 The neutrinos the draft declares but does not use

[1] assigns (ν_{1R}, ν_{2R}) to a **2** and ν_{SR} to a **1**, and then writes no neutrino Yukawa or mass term. These fields are not inert. `feynlag` enumerates five further S_3- and gauge-invariant Dirac structures through $\tilde{H}$, the same pattern as the charged leptons. Majorana ν_R masses would come on top of these; they were not enumerated. The charged-lepton sector is complete; the neutrino sector is simply absent.

8 Open questions, and what would help

1. **State $\mu_5 = \mu_4$ as an assumption in the draft** (§4). It is the $O_L = O_R$ ansatz. The masses alone do not force it.
2. **A complex-Yukawa extension.** The soft-broken fit matches the three CKM magnitudes, but $J = 0$ identically. Whether complex Yukawas can also reproduce $\delta_{\rm CP}$ is open.
3. **The neutrino sector**: the Dirac Yukawas through $\tilde{H}$ and the Majorana ν_R masses. `feynlag` has the seesaw machinery, so this is a matter of doing it.
4. **Does the quark structure belong in the LFV paper?** The CKM obstruction and the LFV structure come from the same O_{12}, and $\mathcal{B}(h \to \tau e) = 0$ exactly in the decays report is the leptonic face of $V_{us} = 0$. My instinct is that a short section is worth it. Your call.

A The derivation ledger

The full typeset version is the companion document `derivations_02.tex`; in summary:

claim	status	independent checks
$\mu_k = Y_k v/\sqrt{2}$, and the draft's uniform factor $\sqrt{2}$	derived	read off the extracted mass matrix; all five ratios equal; rebuilding with the draft's own prefactors reproduces its printed values exactly
$(\sigma_1-\sigma_2)^2 = (\rho-\mu_3)^2+4(\mu_4+\mu_5)^2$	derived	built from $\mathrm{tr}(A^{\mathsf{T}}A)$ and $\det A$; symmetric under $\mu_4 \leftrightarrow \mu_5$; the printed form is shown to be the eigenvalue gap
First-generation decoupling iff $\|r\| = \sqrt{3}$, hence $V_{us} = 0$	derived	eigenvector taken from `eigenvects()`, not written down; isolated root; numeric $V_{\rm CKM}$ at arbitrary Yukawas
The S_3 lepton assignment and its five structures	**imported** [1]	all five invariant; an independent enumeration finds exactly five

References

[1] M. Zeleny-Mora, M. Mondragón and T. A. Valencia-Pérez, *Exploring LFV Higgs decays in the Three Higgs Doublet Model*, draft (`paper_lfvhd/LFVHD_3HDMS3.tex`).

[2] D. Das, U. K. Dey and P. B. Pal, *S_3 symmetry and the quark mixing matrix*, Phys. Lett. B **753**, 315 (2016), arXiv:1507.06509, doi:10.1016/j.physletb.2015.12.038.

[3] S. Navas *et al.* (Particle Data Group), *Review of Particle Physics*, Phys. Rev. D **110**, 030001 (2024), doi:10.1103/PhysRevD.110.030001.